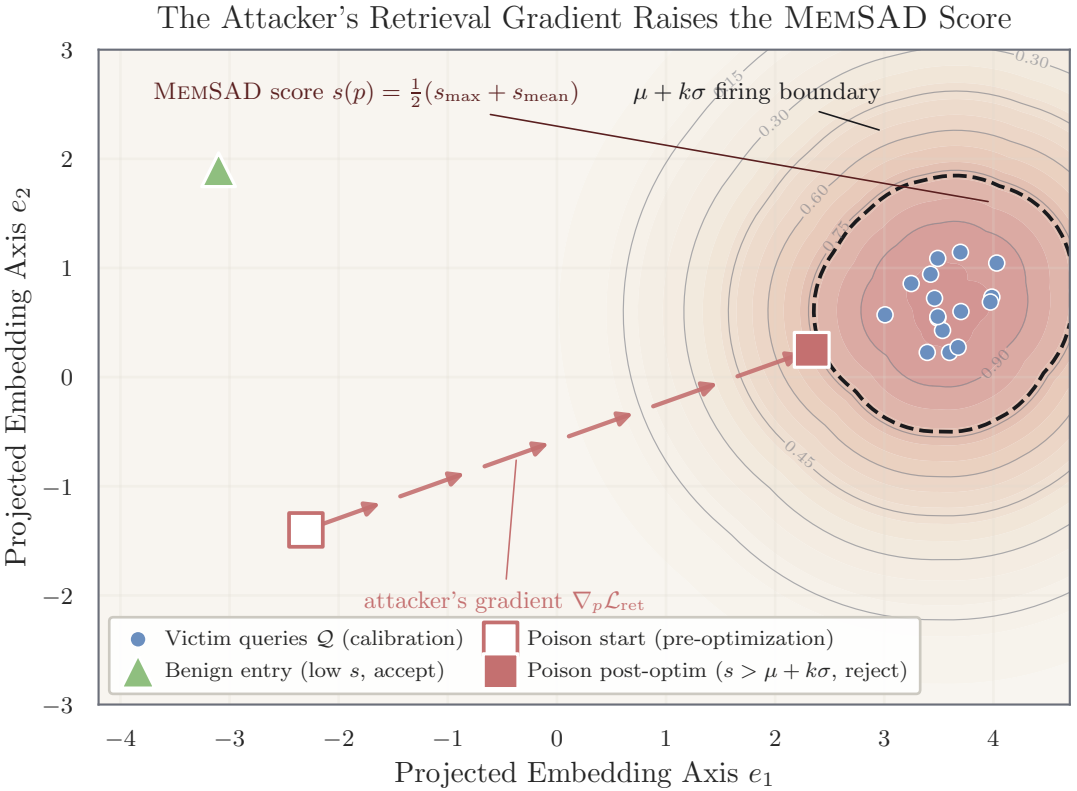
The Attacker's Retrieval Gradient Raises the MemSAD Score
MemSAD score $s(p) = \frac{1}{2}(s_{\max} + s_{\text{mean}})$
$\mu + k\sigma$ firing boundary
attacker's gradient $\nabla_p \mathcal{L}_{\text{ret}}$
0.15
0.30
0.45
0.60
0.75
0.90
Victim queries $\mathcal{Q}$ (calibration)
Benign entry (low s, accept)
Poison start (pre-optimization)
Poison post-optim ($s > \mu + k\sigma$, reject)
Projected Embedding Axis e_1
Projected Embedding Axis e_2
−4
−3
−2
−1
0
1
2
3
4